



Amazon Sales Analysis

Data Science Project using Python

Presented by: Gaurvi Thakur

PYTHON

PANDAS

NUMPY

MATPLOTLIB

SEABORN

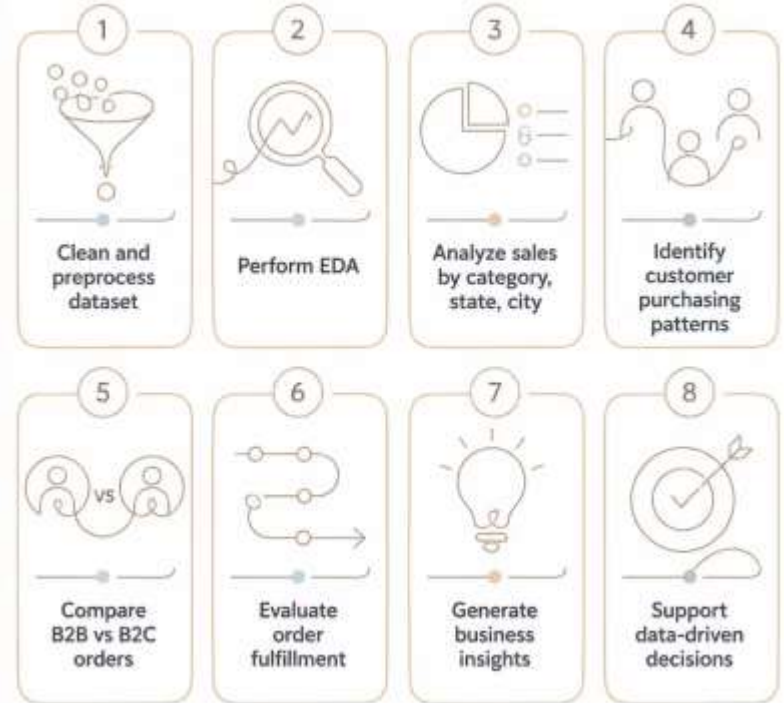
JUPYTER NOTEBOOK

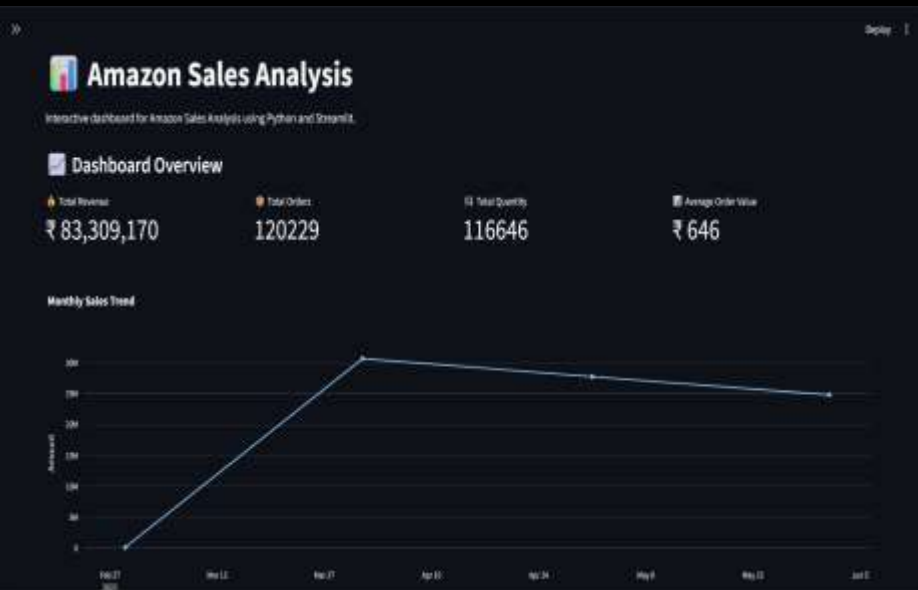
STREAMLIT

GitHub

Problem Statement & Objectives

Amazon generates massive sales data across categories, cities, and states. Manual analysis is slow and error-prone — this project uses **Python-based EDA** to transform raw data into actionable insights.





Project Overview

Analyze Amazon sales data to uncover patterns, trends, and actionable business insights.



Sales Performance

Revenue trends over time



Customer Patterns

Purchasing behavior analysis



Category Performance

Top-selling product segments



Regional Distribution

State & city-level sales

Dataset Overview

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
Index	Order ID	Date	Status	Fulfillment	Sales Channel	Ship-Service	Category	Size	Counter	Qty	Currency	Amount	Ship-City	Ship-State	Ship-Posta	Ship-Count	B2B	Fulfilled-by
1	0 405-80781	2019-01-01	Canceled	Merchant	Amazon.in Standard	T-shirt	S	On the We	0	INR	847.62	MUMBAI	MAHARASH	400081	IN	FALSE	Easy Ship	
2	1 175-91581	2019-01-01	Shipped	Merchant	Amazon.in Standard	Shirt	XL	Shipped	1	INR	406	BENGALURU	KARNATAKA	560005	IN	FALSE	Easy Ship	
3	2 404-09878	2019-01-01	Shipped	Amazon	Amazon.in Expedited	Shirt	XL	Shipped	3	INR	129	RAIPUR	MAHARASH	451211	IN	FALSE		
4	3 403-88152	2019-01-01	Canceled	Merchant	Amazon.in Standard	Blazer	XL	On the We	0	INR	753.33	PUDUCHI	PUDUCHI	605006	IN	FALSE	Easy Ship	
5	4 407-10001	2019-01-01	Shipped	Amazon	Amazon.in Expedited	Theross	XL	Shipped	1	INR	374	CHENNAI	TAMIL NADU	600073	IN	FALSE		
6	5 404-34605	2019-01-01	Shipped	Amazon	Amazon.in Expedited	T-shirt	XL	Shipped	1	INR	324	GHAZIABAD	UTTAR PRA	201102	IN	FALSE		
7	6 408-57484	2019-01-01	Shipped	Amazon	Amazon.in Expedited	T-shirt	L	Shipped	1	INR	853	CHANDIGARH	CHANDIGARH	160006	IN	FALSE		
8	7 406-78077	2019-01-01	Shipped	Merchant	Amazon.in Standard	Shirt	S	Shipped	1	INR	399	HYDERABAD	TELANGANA	500003	IN	FALSE	Easy Ship	
9	8 407-54430	2019-01-01	Canceled	Amazon	Amazon.in Expedited	T-shirt	XL	Canceled	0			HYDERABAD	TELANGANA	500008	IN	FALSE		
10	9 403-83953	2019-01-01	Shipped	Amazon	Amazon.in Expedited	Shirt	XL	Shipped	1	INR	393	CHENNAI	TAMIL NADU	600041	IN	FALSE		
11	10 407-56538	2019-01-01	Shipped	Amazon	Amazon.in Expedited	Shirt	S	Shipped	1	INR	485	CHENNAI	TAMIL NADU	600073	IN	FALSE		
12	11 175-80384	2019-01-01	Shipped	Amazon	Amazon.in Expedited	Shirt	XL	Shipped	3	INR	364	KOTA	UTTAR PRA	301303	IN	FALSE		
13	12 405-55138	2019-01-01	Shipped	Merchant	Amazon.in Standard	Shirt	XS	Shipped	1	INR	399	Anandavil	MAHARASH	444606	IN	FALSE	Easy Ship	
14	13 408-79558	2019-01-01	Shipped	Amazon	Amazon.in Expedited	T-shirt	XS	Shipped	1	INR	857	MUMBAI	MAHARASH	400053	IN	FALSE		
15	14 408-12980	2019-01-01	Shipped	Merchant	Amazon.in Standard	T-shirt	L	Shipped	1	INR	771	MUMBAI	MAHARASH	400053	IN	FALSE	Easy Ship	
16	15 403-83055	2019-01-01	Shipped	Merchant	Amazon.in Standard	Shirt	XL	Shipped	1	INR	544	GUNTUR	ANDHRA PRA	515803	IN	FALSE	Easy Ship	
17	16 406-91761	2019-01-01	Shipped	Amazon	Amazon.in Expedited	Shirt	XL	Shipped	1	INR	329	JAWAHAR	RAJASTHAN	302003	IN	FALSE		
18	17 405-90138	2019-01-01	Shipped	Amazon	Amazon.in Expedited	Shirt	XL	Shipped	1	INR	399	NEW DELHI	DELHI	110074	IN	FALSE		
19	18 403-80303	2019-01-01	Shipped	Merchant	Amazon.in Standard	Shirt	XL	Shipped	1	INR	498	Gurgaon	HARYANA	122004	IN	FALSE	Easy Ship	
20	19 405-50578	2019-01-01	Shipped	Amazon	Amazon.in Expedited	T-shirt	XS	Shipped	1	INR	886	BENGALURU	KARNATAKA	560017	IN	FALSE		
21	20 405-90077	2019-01-01	Shipped	Amazon	Amazon.in Expedited	T-shirt	XL	Shipped	1	INR	517	TIRUCHEN	TAMIL NADU	620018	IN	FALSE		
22	21 404-34045	2019-01-01	Shipped	Amazon	Amazon.in Expedited	T-shirt	M	Shipped	1	INR	606	BENGALURU	KARNATAKA	560040	IN	FALSE		
23	22 175-12095	2019-01-01	Shipped	Amazon	Amazon.in Expedited	Shirt	L	Shipped	1	INR	376	HYDERABAD	TELANGANA	500072	IN	FALSE		
24	23 404-80188	2019-01-01	Canceled	Merchant	Amazon.in Standard	T-shirt	M	On the We	0	INR	570.68	Jaipur	RAJASTHAN	302004	IN	FALSE	Easy Ship	
25	24 403-33840	2019-01-01	Shipped	Amazon	Amazon.in Expedited	Shirt	L	Shipped	1	INR	499	TEEPUR	ASSAM	784003	IN	FALSE		
26	25 405-81913	2019-01-01	Shipped	Merchant	Amazon.in Standard	T-shirt	M	Shipped	1	INR	542	RANCHI	JHARKHAND	834002	IN	FALSE	Easy Ship	
27	26 403-82584	2019-01-01	Shipped	Merchant	Amazon.in Standard	Shirt	XL	Shipped	1	INR	299	DILASPUR	CHHATTISGARH	493003	IN	FALSE	Easy Ship	
28	27 408-34843	2019-01-01	Shipped	Amazon	Amazon.in Expedited	Shirt	XL	Shipped	1	INR	499	Jaipur	RAJASTHAN	302003	IN	FALSE		
29	28 405-77524	2019-01-01	Shipped	Amazon	Amazon.in Expedited	Shirt	M	Shipped	1	INR	399	BENGALURU	KARNATAKA	560037	IN	FALSE		
30	29 404-95538	2019-01-01	Canceled	Merchant	Amazon.in Standard	Shirt	XL	On the We	0			GIRIDHAR	ASSAM	781003	IN	FALSE	Easy Ship	

Key Features: Order ID, Date, Category, Status, Amount, Quantity, Size, State, City, Fulfilment, Sales Channel, B2B

 Clean dataset ready for analysis after preprocessing

128,976

Total Records

21

Features

Project Workflow



Data Collection



Data Cleaning



Preprocessing



EDA



Visualization



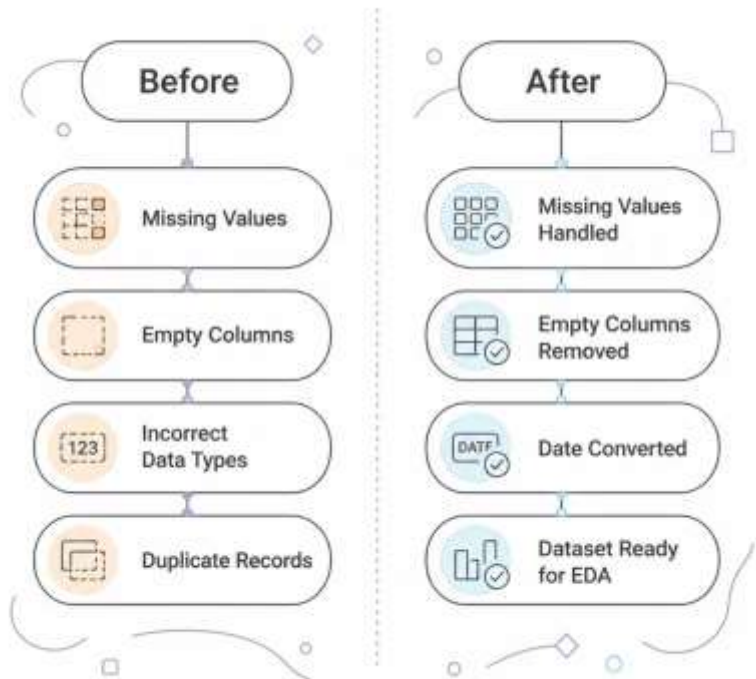
Business Insights



Conclusion

End-to-end pipeline from raw data to actionable insights using Python.

Data Preparation



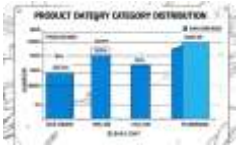
Cleaning Steps Applied

- Removed null & duplicate entries
- Dropped irrelevant columns
- Converted date formats
- Standardized data types

🔄 Dataset validated & ready for EDA

Exploratory Data Analysis

Python-generated visualizations reveal key patterns in the Amazon sales dataset.



Category Distribution

Bar chart — top product categories by sales volume



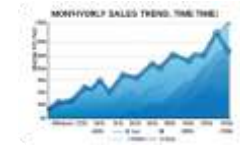
Order Status

Pie chart — completed, pending & cancelled orders



Sales Amount

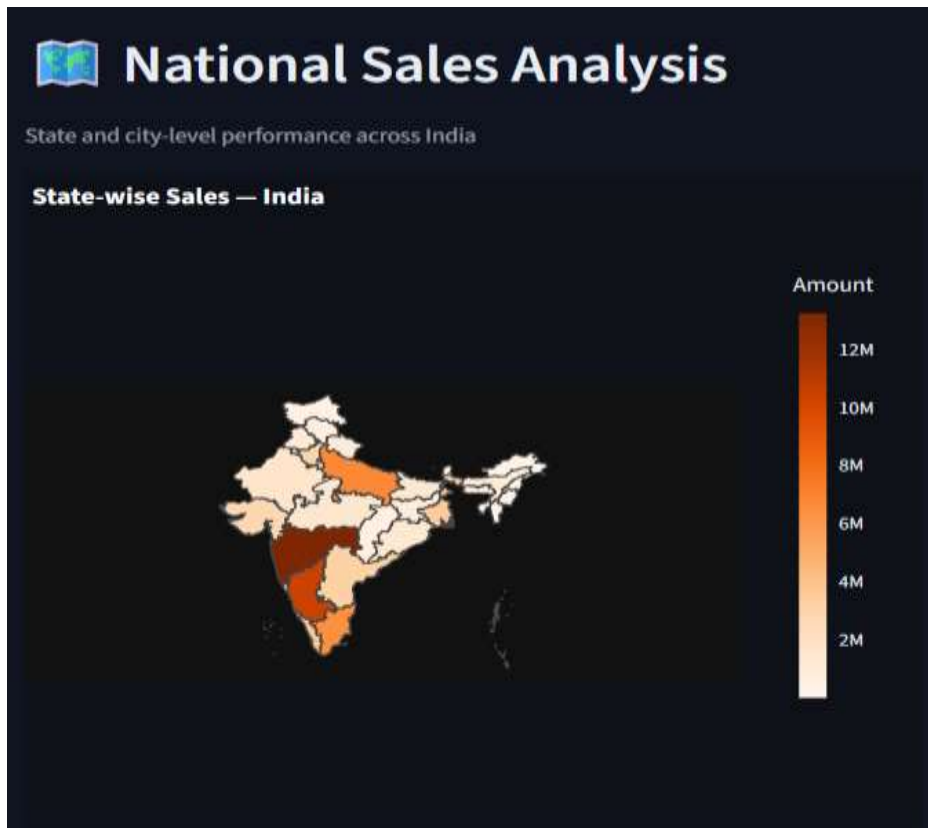
Box plot — spread, outliers & median sales values



Monthly Sales Trend

Line chart — revenue progression across months

Location-wise Sales Analysis (India)



📍 Top 10 States by Sales

Bar chart placeholder — highest-revenue states

🏙️ Top 10 Cities by Orders

Bar chart placeholder — most active cities

🌍 Regional Distribution

Map visualization — geographic sales spread

📌 Key insight: Concentrated sales in specific regions reveals expansion opportunities

Key Business Insights



Top Category

Highest-selling product segment identified



Best State

Leading state by total sales revenue



Top Size

Most ordered product size across categories



Order Status

Majority of orders successfully completed



Sales Trend

Monthly revenue pattern & seasonality



Customer Behavior

B2B vs B2C purchasing patterns

Pros, Cons & Future Scope

✓ Advantages

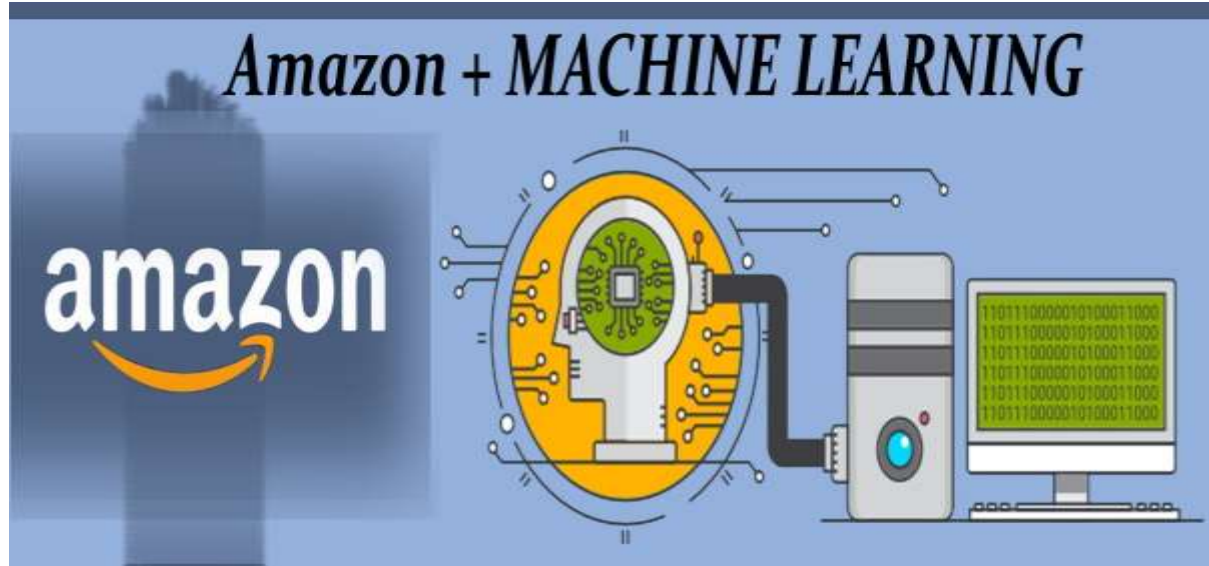
- Clear sales performance view
- Supports data-driven decisions
- Identifies customer trends

⚠ Limitations

- Limited time period
- No profit/cost data
- Missing demographics

🚀 Future Scope

- Sales forecasting models
- Power BI dashboard
- ML recommendation system





Conclusion

01

Data Cleaning

Dataset validated & preprocessed

02

EDA & Visualization

Key sales patterns uncovered

03

Location-wise Analysis

Regional insights mapped

04

Business Insights

Actionable recommendations delivered

Thank You